

Eslam Ashraf Elsawy

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SUMMARY

AI & Cloud Engineer specializing in designing and building production-ready Generative AI applications using LLMs, RAG, and Microsoft Azure. Experienced in developing scalable backend systems, integrating AI with cloud services and external platforms, and delivering end-to-end AI solutions from architecture to deployment. Passionate about building reliable, enterprise-grade AI products with a strong focus on performance, scalability, and practical business impact.

EDUCATION

Faculty of Computers and Artificial Intelligence, Benha University

Benha, Egypt

Bachelor's Degree in Computer Science – Very Good with Honors

Sep 2018 – Jul 2022

Digital Egypt Builders Initiative (DEBI), Ministry of Communications

Egypt

Diploma in AI, Data Science, and Cloud Computing

Jan 2024 – Dec 2024

WORK EXPERIENCE

AI & Cloud Engineer

Remote

Konsultrix

Mar 2026 – Present

- Designed and developed enterprise AI applications using Python, FastAPI, and Azure cloud services.
- Built and enhanced production-ready RAG systems with document processing, semantic search, conversation memory, and guardrails.
- Designed backend architecture, REST APIs, database schemas, and vector search pipelines using MongoDB and Qdrant.
- Integrated AI solutions with Microsoft Teams, Meta platforms, and external cloud services through secure APIs and webhooks.
- Implemented advanced LLM capabilities including prompt engineering, structured outputs, evaluation pipelines, and agent-based workflows.
- Leveraged Azure OpenAI, Azure AI Foundry, Azure Monitor, Azure Functions, App Services, Key Vault, Managed Identity, and Azure DevOps.
- Contributed to technical architecture decisions by evaluating scalability, performance, cloud services, and cost trade-offs for AI solutions.

AI Engineer (Freelance)

Remote

Independent Contractor

Dec 2024 – Mar 2026

- End-to-end AI solutions for tourism and enterprise clients, managing full project lifecycle from requirements gathering to deployment.
- Backend systems using FastAPI and Redis.
- Conversational AI with Azure Cognitive Services (Text Analytics, Translator, Speech) for Arabic (Egyptian dialect) and English.
- Meta Integrated omnichannel messaging platforms (WhatsApp, Facebook Messenger, Instagram) using Meta Graph API webhooks for real-time customer engagement.
- Applied LLM fine-tuning techniques (Llama 3) and implemented custom STT/TTS pipelines optimized for Arabic language processing.

AI Developer (Team Lead) – EduVision AI Platform

Remote, Egypt

Ministry of Communications and Information Technology (MCIT)

Nov 2024 – Mar 2025

- Led cross-functional team to design and implement a government digital transformation solution, coordinating technical development and stakeholder communication.
- Managed end-to-end ML pipeline from data collection to Azure Cloud deployment, ensuring on-time delivery for ministerial demonstration.
- Applied multimodal fusion (video, audio, text), model fine-tuning, and performance optimization, reducing processing time by 40%.
- Contributed to an AI solution recognized by MCIT as a promising startup prototype.

AI Engineer (Internship – Team Lead)

National Bank of Egypt (NBE)

Remote, Egypt

July 2024 – Oct 2024

- Led development team to create a AI-powered analytics tool for financial data analysis, managing technical implementation and quality assurance.
- Designed hybrid AI architecture combining computer vision and NLP for automated data extraction and intelligent query processing.
- Implemented voice-enabled features with Azure TTS supporting bilingual output (Arabic/English) for enhanced user accessibility.
- Delivered desktop application deployed across banking terminals, reducing analyst review time by 60%.

PROJECTS

- **EduVision: Multimodal AI Performance Evaluation System** - MCIT [\[View\]](#)
 - Led development of multimodal AI system analyzing video, audio, and text to automate HR performance reports for the Ministry of Communications.
 - Built pipeline integrating OpenPose, MediaPipe, Whisper, and fine-tuned Llama 3 / Qwen-VL, achieving 90%+ accuracy across 100+ sessions.
 - Optimized Azure deployment and model selection, cutting processing time by 40% and cloud costs by 30%; recognized by MCIT as an innovative startup prototype.
- **Smart Tourism Assistant: RAG-Powered Omnichannel Chatbot**
 - Built RAG-powered chatbot backend handling multi-turn conversations across WhatsApp, Messenger, Instagram, and Web platforms.
 - Implemented ChromaDB retrieval with HuggingFace reranking (92% accuracy) and Gemini 2.5 Flash for Egyptian Arabic intent detection.
 - Delivered 95% user satisfaction and 80% faster response times, autonomously resolving 70% of customer inquiries.
- **Financial Dashboard Auto Transcript Analyzer** - NBE [\[View\]](#)
 - Led development of Gen-AI analytics tool for National Bank of Egypt, automating financial report generation by extracting data from 500+ financial chart images using Vision Transformers (Donut) with 85%+ accuracy.
 - Architected RAG-inspired Q&A system combining FAISS vector search with GPT-NeoX-20B for contextual financial analysis.
 - Integrated Azure TTS and Translator for bilingual voice output (Arabic/English); deployed FastAPI microservice with PyQt5 desktop client, reducing analyst review time by 60%.

CERTIFICATES

- **Microsoft Certified: Azure AI Engineer Associate (AI-102)** – Earned Aug 2025 [\[View\]](#).

TECHNICAL SKILLS

Languages: Python (Advanced), C++ , Java.

Generative AI & LLMs: OpenAI Models, Google Gemini, Meta Llama, Anthropic Models, LangChain, RAG, Prompt Engineering, Agentic AI, MCP.

Backend: FastAPI, REST APIs, Async Programming, Redis, Celery, WebSockets, Webhooks.

Cloud: Microsoft Azure (OpenAI, AI Foundry, App Services, Functions, Storage, Monitor, Managed Identity, Key Vault, Container Apps, Azure DevOps).

Vector DBs: Qdrant, ChromaDB, FAISS, Pinecone.

Databases: MongoDB, PostgreSQL, MySQL.

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Transformers, CNNs, RNNs.

NLP: Hugging Face, spaCy, NLTK, NER, Text Classification, Sentiment Analysis.

Computer Vision: OpenCV, MediaPipe, YOLO, Vision Transformers.

Speech: Whisper, Azure Speech, Azure Neural TTS, Audio Processing.

Integrations: Microsoft Teams, Microsoft Graph API, Meta Graph API, Webhooks.

DevOps: Docker, Git, GitHub, CI/CD, PowerShell, Bash.

Frontend: React, Streamlit, PyQt5.

ERP: Odoo API Integration, Custom Module Development.